QDP-1: A Multi-Stage Framework for Detecting Quantum Spin Coherence in Radical Pair Systems

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Abstract

We propose QDP-1, a seven-stage detection protocol for identifying quantum spin coherence effects in radical pair systems under ambient conditions. The framework makes two methodological contributions: (1) it unifies angular dependence testing, field-strength power-law discrimination, temperature-dependent coherence scaling, and negative control gating into a single scored detection framework for radical pair systems; and (2) it proposes adapting zero-noise extrapolation (ZNE), a proven error-mitigation technique from quantum computing, as a diagnostic fingerprint for distinguishing coherent spin dynamics from classical behavior–though this component remains a hypothesis awaiting theoretical grounding and experimental validation. The protocol borrows structural elements from particle physics discovery criteria (5σ thresholds, control gating) and clinical trial methodology (staged evidence, artifact elimination). We validate QDP-1 in simulation across four radical pair candidate systems and one known-negative control (a non-radical-pair fluorophore), demonstrating that the protocol differentiates between systems and that the control gate correctly caps verdicts when artifact contamination is present. The ZNE diagnostic (Stage 5) is proposed but not yet validated: the simulation's decoherence proxy does not produce the exponential decay signature the stage is designed to detect, and experimental validation with physical decoherence mechanisms is required. No existing framework in the quantum biology literature unifies these elements into a single system-agnostic protocol. QDP-1 is validated in simulation only; experimental validation is needed. Simulation code is available at `https://github.com/HighpassStudio/qdp1`.

I. INTRODUCTION

The radical pair mechanism is the leading candidate explanation for biological magnetoreception and has been confirmed in cryptochrome systems through fluorescence-detected magnetic field effects [1] and Earth-strength magnetic sensitivity in tightly bound radical pairs [2]. Experimental techniques for detecting radical pair quantum effects have matured considerably: magneto-fluorescence fluctuation microspectroscopy enables investigation of quantum effects in biology at the single-molecule scale [3], and sensing magnetic-field effects at the scale of small molecular ensembles is now achievable [4].

Despite this progress, the field lacks a unified detection framework. Individual techniques–fluorescence-based magnetic field effect measurement, angular dependence characterization,

$B_{1/2}$ field-strength analysis, temperature-dependent coherence studies–are applied independently and evaluated by different criteria across different laboratories. This creates several problems:

1. **No standard for sufficiency.** How many positive indicators constitute evidence for quantum spin coherence? There is no consensus threshold analogous to the 5σ standard in particle physics.

2. **No control gating.** Negative controls are used, but there is no formal mechanism to prevent false positives from propagating when controls show contamination.

3. **No cross-disciplinary diagnostic transfer.** Zero-noise extrapolation (ZNE), a technique developed for quantum computing error mitigation [5, 6], provides a natural framework for probing quantum coherence effects through deliberate noise scaling. This approach has begun migrating from quantum computing into quantum sensing [7] but has not been applied to quantum biology or radical pair systems.

4. **No system-agnostic protocol.** Each new radical pair candidate is evaluated with ad hoc methods. A reusable detection kit would accelerate the field.

QDP-1 addresses these gaps by combining seven partially independent tests into a scored verdict with explicit control gating. The protocol is designed to be applied identically to any radical pair system, with each application constituting an independent experimental contribution. This work is intended as a starting point for community evaluation and refinement.

A. Scope

QDP-1 detects signatures *consistent with* quantum spin coherence in radical pair systems. It does not test for entanglement specifically, nor does it address other quantum biological phenomena (coherent energy transfer, quantum tunneling) without additional stages. The framework provides evidence grades, not binary proof.

II. RELATED WORK

A. Radical Pair Detection Methods

The radical pair mechanism and its role in biological magnetoreception have been extensively reviewed [8, 9, 17]. Key experimental milestones include the detection of magnetic field effects on cryptochrome fluorescence [1], characterization of angular dependence in singlet yield [10], measurement of $B_{1/2}$ parameters for field-strength characterization [11], and investigation of temperature effects on spin coherence [12]. These techniques are individually well-established and are not claimed as novel contributions of this work.

B. Zero-Noise Extrapolation in Quantum Computing

ZNE was introduced as an error-mitigation technique for near-term quantum computers [5, 6]. The core idea is to deliberately amplify noise, measure the observable at multiple noise levels, and extrapolate to the zero-noise limit. ZNE is now standard practice in quantum computing and has begun migrating into quantum sensing–notably, a 2024 study applied ZNE-style extrapolation to mitigate errors in DC magnetometry [7]. However, the use of ZNE as a *diagnostic* (to distinguish quantum from classical behavior based on decay curve shape) rather than as an error-mitigation tool, and its application to biological or chemical radical pair systems, does not appear in the mainstream literature.

C. The Quantum Biology Detection Gap

The question "How quantum is radical pair magnetoreception?" has been addressed theoretically [13], establishing that radical pair magnetoreception satisfies formal criteria for quantum behavior from a quantum information perspective. However, this assessment is conceptual, not operational–it does not provide a staged detection protocol for experimental use.

The most relevant methodological advances are the 2024 magneto-fluorescence fluctuation microspectroscopy technique [3] and the study on sensing magnetic-field effects at small ensemble scales [4]. Both move toward reusable measurement methodology for quantum biology, but neither constitutes a multi-stage scored framework with control gating and

cross-disciplinary diagnostic integration.

A 2024 study on simulating spin biology using a digital quantum computer [14] represents the most direct bridge between quantum computing and quantum biology in the existing literature. However, the connection there is different: the paper uses extrapolation ideas akin to ZNE for simulation accuracy, not as an experimental diagnostic for biological systems.

D. Novelty Claim

QDP-1 unifies ZNE-style noise scaling (adapted from quantum computing), Fourier angular power analysis (adapted from signal processing), power-law field discrimination, temperature coherence scaling, and negative control gating (adapted from particle physics and clinical trials) into a single scored detection framework for quantum biology. Based on a search across 2005–2025 literature in major journals and arXiv, no mainstream precedent was found for this full combination. The likely novelty is the assembly, not the individual parts.

III. METHODS: THE QDP-1 PROTOCOL

QDP-1 consists of seven stages, each contributing a weighted score to a final verdict. Stage 3 (negative controls) serves as a gate: if controls fail, the final verdict is capped regardless of total score. The protocol requires a fluorescence-based readout of radical pair reaction yield as a function of applied magnetic field angle and strength.

A. Stage 1: Baseline Noise Characterization (1 point)

Purpose. Determine whether the measurement setup is shot-noise-limited before signal measurement begins.

Method. Measure N fluorescence pulses with no applied magnetic field. Compute the excess noise ratio:

$$\mathrm{ENR} = \frac{\sigma_{\mathrm{meas}}}{\sqrt{\mu}} \tag{1}$$

where μ is the mean detected counts and σ_{meas} is the measured standard deviation.

Decision. ENR < 1.5 passes (shot-noise-limited). ENR ≥ 1.5 flags technical noise dominance.

Scoring. 1 point if passed.

B. Stage 2: Differential Signal Detection (2 points)

Purpose. Detect angular dependence of radical pair yield using normalized differential measurement that cancels common-mode drift.

Method. Block-randomized alternation between field angle $\theta_A = 0$ and $\theta_B = \pi/2$. For each block of $2B$ pulses:

$$C_i = \frac{S_A - S_B}{S_A + S_B} \tag{2}$$

where S_A, S_B are summed photon counts at each angle.

Statistics. Mean contrast $\bar{C} = \langle C_i \rangle$, standard error SE $= \mathrm{std}(C_i)/\sqrt{N_{\mathrm{blocks}}}$, Z-score $Z = |\bar{C}|/\mathrm{SE}$.

Decision. $Z > 5$ for detection (5σ, matching the particle physics discovery threshold). $Z > 3$ for evidence.

Scoring. 2 points if $Z > 5$; 1 point if $Z > 3$.

C. Stage 3: Negative Controls–GATE (2 points)

Purpose. Eliminate artifacts. This stage is a gate: if controls fail, the final verdict is capped at "INCONCLUSIVE" regardless of total score.

Controls.

- C1: Same angle ($\theta_A = \theta_B = 0$). Tests for detector asymmetry.
- C2: No applied field / near-zero field. Tests for field-independent artifacts.
- C3: Killed radical pair (radical scavenger or denatured protein). Tests whether the signal requires the spin-correlated radical pair.

Decision. Each control must show $Z < 2.0$. Any control with $Z > 3.0$ flags artifact contamination.

Scoring. 2 points if all controls pass. 0 points if any control fails. Gate: failure caps verdict.

D. Stage 4: Angular Shape via Fourier Decomposition (2 points)

Purpose. Confirm that the angular dependence follows the $\cos^2\theta$ pattern predicted by radical pair theory.

Method. Measure yield at $N \geq 24$ equally-spaced angles from 0 to π. Compute the Fourier power in the 2θ component:

$$a_2 = \frac{2}{N}\sum_i y_i \cos(2\theta_i) \tag{3}$$

$$b_2 = \frac{2}{N}\sum_i y_i \sin(2\theta_i) \tag{4}$$

$$P_2 = a_2^2 + b_2^2 \tag{5}$$

Since $\cos^2\theta = 0.5 + 0.5\cos(2\theta)$, the radical pair signal concentrates power in the 2θ Fourier component.

Decision. Power ratio P_2/total variance > 0.5 AND $F > 10$ for strong detection.

Scoring. 2 points if both criteria met. 1 point if either met alone.

E. Stage 5: Noise Decay Fingerprint–ZNE Diagnostic (1 point)

Purpose. Distinguish quantum coherence effects from classical artifacts using the shape of signal decay under controlled decoherence.

Method. Deliberately increase decoherence by controlled means (paramagnetic ions, temperature increase, or chemical modification). Measure differential contrast at decoherence levels $d = 0.0, 0.1, \ldots, 0.9$. Fit two models:

$$\text{Linear:} \quad C(d) = a - b \cdot d \tag{6}$$

$$\text{Exponential:} \quad C(d) = a \cdot \exp(-k \cdot d) \tag{7}$$

Rationale. This adapts ZNE from quantum computing [5, 6]. In standard ZNE, noise is deliberately amplified and the observable extrapolated to zero noise. Here, we repurpose the technique as a *diagnostic*: the shape of the decay curve under increasing decoherence distinguishes quantum from classical behavior.

Important caveat. Exponential decay alone is not proof of quantum coherence. Some classical processes (photobleaching, thermal relaxation) also produce exponential decay.

This stage provides supporting evidence, not standalone proof. The scoring weight (1 point of 10) reflects this limitation.

Decision. $R^2_{\text{exp}} - R^2_{\text{lin}} > 0.1$ indicates exponential (quantum-consistent) decay.

Scoring. 1 point if exponential decay shape detected.

1. Theoretical Basis

Under Markovian dephasing modeled via Lindblad operators, the off-diagonal elements of the spin density matrix responsible for singlet-triplet coherence decay as $\exp(-\gamma t)$, where γ is the dephasing rate. When integrated against the pair survival probability $\exp(-k_{\text{eff}} \cdot t)$, the time-averaged coherent modulation amplitude takes a Lorentzian-like form:

$$A(\gamma) \sim \frac{k_{\text{eff}}}{k_{\text{eff}} + \gamma} \tag{8}$$

Since the controlled decoherence parameter d scales the physical dephasing rate linearly–$\gamma(d) = \gamma_0 + c \cdot d$–the contrast follows:

$$C(d) \approx C_0 \cdot \exp(-\beta d) \tag{9}$$

where $\beta = \alpha c$. For classical artifacts, the contrast typically scales proportionally: $C(d) \approx C_0 - b \cdot d$ (linear).

2. Extension: Stretched Exponential for Heterogeneous Systems

Real biological radical pairs embedded in proteins experience heterogeneous environments that produce a *distribution* of dephasing rates. When the rate distribution $\rho(u)$ is broad, the ensemble-averaged contrast becomes:

$$\frac{C(d)}{C_0} = \int_0^\infty \rho(u) \exp(-u \cdot d)\, du \tag{10}$$

which is the Laplace transform of $\rho(u)$. For many disordered systems, this produces the Kohlrausch-Williams-Watts (KWW) stretched exponential:

$$C(d) = C_0 \cdot \exp\left[-\left(\frac{d}{\tau}\right)^\beta\right] \quad \text{where } 0 < \beta \leq 1 \tag{11}$$

The stretching exponent β encodes environmental heterogeneity: $\beta = 1$ corresponds to homogeneous Markovian dephasing; $0.5 < \beta < 1$ indicates heterogeneous or non-Markovian

decoherence with distributed character (quantitative predictions yield $\beta \approx 0.7$–0.9 in certain radical pair regimes); linear decay is consistent with classical artifacts.

Proposed refinement. Fit three models (linear, exponential, stretched exponential) using AIC or BIC for model selection. A stretched exponential with $\beta \in [0.6, 1.0]$ would be classified as "quantum-consistent."

F. Stage 6: Field-Strength Power Law (1 point)

Purpose. Distinguish quantum radical pair response from classical magnetic effects.

Method. Measure differential contrast at field strengths spanning ≥ 2 orders of magnitude. Fit power law to the weak-field regime:

$$\log(C) = \alpha \cdot \log(B) + \beta \tag{12}$$

Rationale. Quantum radical pair theory predicts $\alpha \approx 1$ at weak fields [11]. Classical effects scale as $\alpha \approx 2$.

Decision. $\alpha < 1.5$ indicates quantum-consistent scaling.

Scoring. 1 point if quantum-consistent scaling detected.

G. Stage 7: Temperature Coherence Scaling (1 point)

Purpose. Quantum spin coherence degrades with increasing temperature. Classical magnetic effects typically show different temperature dependence.

Method. Measure differential contrast at temperature offsets from baseline (-10 K to $+30$ K). Compute Spearman rank correlation between temperature and contrast magnitude.

Decision. Spearman $\rho < -0.5$ indicates quantum-consistent thermal scaling.

Scoring. 1 point if significant negative correlation detected.

TABLE I. QDP-1 scoring allocation.

Stage	Test	Max points
1	Baseline noise quality	1
2	Differential signal detection	2
3	Negative controls (GATE)	2
4	Fourier angular shape	2
5	Noise decay fingerprint (ZNE)	1
6	Field-strength power law	1
7	Temperature coherence scaling	1
	Total	**10**

IV. SCORING AND VERDICT

A. Score Allocation

B. Verdict Thresholds

$\geq 8/10$: Consistent with quantum spin coherence. $\geq 6/10$: Strong evidence–recommend independent replication. $\geq 4/10$: Suggestive–additional data needed. $< 4/10$: No detection.

C. Stage Independence

Stages 1–4 provide substantially independent lines of evidence. Stages 5–7 are partially correlated: temperature (Stage 7) affects decoherence rates (Stage 5), and decoherence affects field-scaling behavior (Stage 6). The scoring weights reflect this asymmetry: Stages 1–4 carry 7 of 10 points, while Stages 5–7 carry 3 points combined.

D. Control Gate

Stage 3 must pass for any positive verdict. If controls fail, the verdict is capped at "INCONCLUSIVE–ARTIFACT RISK" regardless of total score. This mechanism is adapted from blind-analysis protocols in particle physics and clinical trial gating.

V. SIMULATION VALIDATION

A. Systems Tested

QDP-1 was validated in simulation across four radical pair candidate systems and one known-negative control:

1. **Avian Cryptochrome Cry4a**–field ratio 0.05, 1000 photons/pulse.
2. **Flavin-Tryptophan Model**–field ratio 0.05, 2000 photons/pulse.
3. **Plant Cryptochrome Cry1**–field ratio 0.05, 800 photons/pulse.
4. **Enzyme Radical Pair**–field ratio 0.03, 500 photons/pulse.
5. **Rhodamine B (Negative Control)**–field ratio 0.0, 1500 photons/pulse.

B. Noise Model

Simulations include five realistic noise sources: Poisson shot noise, LED intensity drift ($\sigma = 0.5\%$), dark counts (20 counts/pulse), photobleaching (exponential decay, rate 0.1), and Gaussian detector read noise ($\sigma = 5$ counts).

C. Results

TABLE II. QDP-1 simulation results (representative run).

System	Score	Verdict
Avian Cryptochrome Cry4a	7/10	Strong evidence
Flavin-Tryptophan Model	5/10	Suggestive
Plant Cryptochrome Cry1	7/10	Strong evidence
Enzyme Radical Pair	9/10	Consistent
Rhodamine B (Control)	3–4/10	No detection

Key observations: (1) Stage 2 cleanly separates RP from non-RP systems ($Z = 17.8$–95.0 for candidates vs. $Z = 0.4$ for control). (2) The negative control scores at the bottom (3–4/10). (3) Stage 5 (ZNE diagnostic) scores 0/1 for all systems–the decoherence proxy does not produce the expected exponential decay. (4) Scores vary ±1–2 points between runs due to stochastic noise.

D. Simulation Limitations

The simulation uses a simplified decoherence model: linear field-ratio reduction as proxy for decoherence, uniform temperature coefficient, and Lorentzian-derived power-law scaling. The simulation validates *protocol logic*, not *physical predictions*.

VI. DISCUSSION

A. The Cross-Disciplinary Bridge

QDP-1 sits at the intersection of quantum computing (ZNE), spin chemistry (radical pair detection), and signal processing (Fourier analysis), importing methodology from particle physics and clinical trials. The contribution is the assembly, not new measurement physics.

B. The ZNE Adaptation: Status and Limitations

The proposed use of ZNE as a diagnostic rather than an error-mitigation tool is QDP-1's most novel element–and its least validated. Stage 5 scores 0/1 across all systems in simulation. Three concerns persist: (1) many classical processes also produce exponential decay; (2) deliberate decoherence perturbation may alter chemistry, not just spin coherence; (3) the theoretical basis, while grounded in Lindblad dynamics (Sec. III E 1), requires validation against full spin-dynamical simulations and experiment.

C. Practical Application

Equipment requirements for optical/magnetic components are modest ($1,500–$4,000). However, biological sample preparation (protein expression, anaerobic handling, EM shield-

ing) adds substantial infrastructure requirements. Priority targets: (1) Cry4a, (2) synthetic FAD-Trp models, (3) plant cryptochromes, (4) enzyme radical pairs.

D. Broader Implications

If validated experimentally, QDP-1 enables: quantum biology surveys across cryptochrome families, pharmacological screening for radical pair involvement [15], and investigation of radical pair mechanisms in oxidative stress and reactive oxygen species production [16].

VII. LIMITATIONS

1. **Simulation only.** Experimental validation needed.
2. **Simplified decoherence.** Stage 5 uses a proxy model.
3. **Uncalibrated scoring weights.** Need empirical validation.
4. **Partial stage correlation.** Stages 5–7 are correlated.
5. **Scope limitation.** Radical pair systems only.
6. **Exponential decay caveat.** Not sufficient alone.
7. **Missing standard tests.** RF disruption, isotope substitution, EPR not included.
8. **Stage 4 false positives.** Fourier test can score on noise-only data.
9. **Run-to-run variability.** $\pm$1–2 points between runs.

VIII. PROPOSED EXTENSIONS

Stage 8: RF Magnetic Field Disruption. RF at the Larmor frequency should disrupt singlet-triplet interconversion [8].

Stage 9: Magnetic Isotope Substitution. Replacing magnetic nuclei (^{1}H $\rightarrow$ ^{2}H, ^{12}C $\rightarrow$ ^{13}C) alters hyperfine coupling predictably.

Stage 10: Direct Radical Characterization via EPR/Transient Absorption. Time-resolved EPR confirms radical identity.

IX. CONCLUSION

QDP-1 provides a structured, reproducible framework for evaluating quantum spin coherence signatures in radical pair systems. Its primary contribution is formalizing multi-stage evidence accumulation with control gating–a standardization that does not currently exist in the field. The proposed ZNE diagnostic represents a speculative but potentially valuable cross-disciplinary transfer requiring further validation.

QDP-1 is a methodological proposal–a structured starting point for standardization–not a battle-tested detection kit. Its value will be determined by experimental validation, integration with established techniques, and collaborative stress-testing by the spin chemistry and quantum biology communities.

We invite collaboration from experimental groups working with radical pair systems.

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Simulation code and full paper are available at `https://github.com/HighpassStudio/qdp1`. This work benefited from AI-assisted analysis and literature review using Claude (Anthropic).

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